Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/29/2015
PI Name:	Natarajan, Mohan Ph.D.		
Project Title:	Targeting Protein Cross Talk Signaling as a Biological Predictor for Space Radiation Cancer Risk		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	natarajan@uthscsa.edu	**Fax:**	FY 210-567-4098
PI Organization Type:	UNIVERSITY	**Phone:**	210-567-5663
Organization Name:	The University of Texas Health Science Center at San Antonio		
PI Address 1:	Pathology		
PI Address 2:	7703 Floyd Curl Dr		
PI Web Page:			
City:	San Antonio	**State:**	TX
Zip Code:	78229-3901	**Congressional District:**	21
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Space Radiobiology NNJ11ZSA001N
Start Date:	04/01/2012	**End Date:**	10/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 10/31/2015 per NSSC information (Ed., 1/5/15)		
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Blakely, Eleanor (Lawrence Berkeley National Laboratory) Mohan, Sumathy (University of Texas, San Antonio) Prihoda, Tom (University of Texas, San Antonio)		
Grant/Contract No.:	NNX12AC32G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	One of the important concerns of 'late effects' after space radiation exposure is the onset of cancers. The onset of cancer could be initiated by radiation-induced molecular changes that are persistently maintained through a positive feed back cycle and amplified over time. We propose that two key proteins, nuclear factor kappa Beta (NF-kB) and tumor necrosis factor alpha (TNF-a), sequentially co-ordinate and control this cascade of events. A novel role of these proteins in radiation-induced tumor development is identified through our recent in vitro and in vivo studies after low linear energy transfer (LET) radiation exposure. In this project we address: (i) how low doses of high LET radiation trigger initial damage signals that remain persistent and are maintained in the memory of the cell/tissue, (ii) how these signals are amplified, (iii) how these amplified signals are transmitted to result in a sustained radial expansion of initial damage, and finally (iv) how radial communication of these signals activates factors that are responsible for predisposing the susceptible cells to carcinogenesis. We will examine these objectives using both in vitro (parallel plate flow model) and ex-vivo (mouse-artery organ culture) approaches, in which irradiated normal vascular endothelial cells in an intact tissue setup or organ culture will be allowed to communicate with the co-existing non-targeted normal or pro-tumorigenic cells from lung, skin, and breast tissue. An in vivo animal model will be used to prove the tumorigenic potentials of transformed cells. We will compare the outcomes with regard to the LETs (51 and 181 keV/um) of two different HZE (high energy transfer) ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/amu) and having similar track structure dimensions, but different ionization densities. This project addresses four of the seven major risks and gaps (Cancer 1, 2, 3, & 4) identified in the Space Radiation risks.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study will clarify whether radiation-induced molecular changes that are persistently maintained through positive feed-back cycle through two key proteins, nuclear factor kappa B (NF-kB) and tumor necrosis factor alpha (TNF-a), are amplified over time. These altered molecular events are responsible for clonal expansion of cells that are prone to be tumorigenic. In terms of usefulness of the above study on Earth, normal tissue response is one of the major focuses to achieve acceptable therapeutic gain in radiotherapy of many cancers. Identification and characterization of the molecular mediators that play a role in radiation-induced bystander effects that could lead to genomic instability and survival advantage after therapeutic doses of low LET radiation is important to understand the radiation-induced delayed effect in normal tissue. Therefore the findings from the study proposed with space radiation can set a stage to understand the underlying mechanism of tumorigenesis, tumor recurrence, and tumor growth at the treatment site.
Task Progress:	One of the important concerns of 'late effects' after space radiation exposure is the onset of cancers. The onset of cancer could be initiated by radiation-induced molecular changes that are persistently maintained through positive feed-back cycle and amplified over time. We hypothesize that a bystander effect through intercellular communication is initiated by TNF-a through activation of NF-kB. While TNF-a and NF-kB can induce one another through positive feedback mechanism and maintain a “memory” of the initial insult for an extended period, secreted TNF-a generates low profile, transmissible genetic damages through rapid production of free radicals and simultaneously triggers NF-kB activation in bystander cells. Development of concurrent survival advantage, which is mediated by the persistent activation of NF-kB could permit the cells to accumulate and carry over the transmissible genetic damage, which is caused by TNF-a, to the next generation. The domino effect caused by this course of bystander mechanism may readily alter the physiology of normal tissue and cause neoplastic transformation and cancer initiation. We proposed to examine the objectives using both in vitro and in vivo approaches. We proposed to compare the outcomes with regard to the LETs (51 and 181 keV/um) of two different HZE ion beams (28Si and 56Fe) accelerated to the same velocity (600 MeV/amu) and having similar track structure dimensions, but different ionization densities. This project addresses four of the seven major risks and gaps (Cancer 1, 2, 3, & 4) identified in the Space Radiation risks. None of the originally proposed specific aims are changed. In the third year funding period, proposed in vitro studies pertaining to the changes in the bystander cells specified in Aim 2 and the read out at the cellular level included under Aim 3 were completed. A minimum of three repeats were carried out in each experiment. In addition complementary key in vivo studies in normal mice exposed to iron ion were also carried out. One manuscript and nine meeting presentations (oral and poster) were achieved. Out of the data presented in the Human Research Program (HRP) and Radiation Research Society (RRS) meetings two manuscript are being prepared. The summary of the experimental results are presented below. First, we examined whether secreted TNF-a after particle radiation could initiate a feedback cycle in the bystander cells similar to the targeted cells to maintain the initial insult for an extended period of time by activating NF-kB in the bystander cells. Primary human aortic endothelial cells in 6-well plates were transfected with luciferase reporter gene with putative binding sites of NF-kB (pNF-kB-Luc) in the promoter region. Cells in the culture inserts exposed to 0.8 Gy of 56Fe (600 MeV/u) particle radiation were co-cultured with the pNF-kB-Luc transfected cells. Cells transfected with pTAL-Luc were used as negative control. Luciferase activity was measured by Bright-Glo™ luciferase assay system (Promega, WI) at 16 h post co-culture period. To specifically identify the role of TNF-a in regulating NF-kB activation in the neighboring cells as part of bystander effect, the transfected cells were incubated with TNFRI blocker 30 min prior to placing the inserts with irradiated cells. The results showed an increased luciferase activity in the cells that are co-cultured with irradiated cells. An increase of 95.81% was observed compared to mock-irradiated control. Whereas pre-treatment with anti-TNFRI blocker, which blocks the receptors on the transfected cells, attenuates this induced luciferase activity by 73.26% from radiation induced NF-kB activity indicating that TNF-a secreted by irradiated cells, in fact, is responsible for the induction of the functional activity of NF-kB in the bystander cells. In the second set of studies, first, we tested whether TNF-a through paracrine mechanism could cause free radical generation in the unirradiated bystander cells. Human aortic endothelial cells (HAECs) grown on gelatin coated glass coverslips placed in a 6-well culture plates and incubated with cells in the inserts that are exposed to 0.8 Gy of 56Fe (600 MeV/u) particle radiation for 3 h were stained with dihydroethidium (DHE, 0.5µM). To confirm the free radicals generated are largely mediated through TNF-a, parallel cultures were incubated with TNFRI blocker 30 min prior to co-culturing the cells. There was significant increase in the free radical production in the bystander cells co-incubated with irradiated cells. This induction was inhibited by 52.5% when the cells are pre-treated with TNFRI blocker, indicating the free radical generation in the bystander cells are mediated through TNF-a secreted from irradiated cells. Second, we investigated whether NF-kB-dependent TNF-a is responsible for causing DNA damage in bystander cells. HAECs either exposed to 0.8 Gy or mock irradiated were co-cultured with non-irradiated cells, both sharing common medium. FITC labeled anti-gamma H2AX antibody was used to detect DNA damage in the bystander cells counter

stained with DAPI at 3 h or 24 h after incubation with irradiated cells. Identical set of experiments were performed in the presence of a TNF-a blocker, TNFRI inhibitor or NF-kB inhibitor, isohelanin. Cells that were mock irradiated or treated with H2O2 (25 μM) for 30 min used as negative and positive controls, respectively. Bystander cells incubated with irradiated cells showed induced foci and those were significantly blocked either by NF-kB blocker or the TNF-a receptor blocker indicating the role of NF-kB-TNF-a cycle in mediating genetic damage in the bystander cells.

In the third set of studies we demonstrate that the activation of NF-kB signaling in the bystander cells is responsible for induced survival advantage. NF-kB-mediated survival advantage was assessed in the co-cultured bystander cells by examining persistent activation of NF-kB-mediated pro survival and anti-apoptotic pathway (BCL-2, MCL-1, cIAP (Birc2), Ki-67, surviving, pGSK3b). The activation of markers that are known to be involved anti-apoptosis and survival advantage were induced when the cells are co-incubated with irradiated cells. Those activations were significantly suppressed in the presence of TNFRI blocker.

The results from the first part of the study revealed that (i) the occurrence of radiation-induced bystander effect; (ii) bystander mechanism involves TNF-a paracrine pathway and activates NF-kB transactivation potentials for down-stream gene transactivation in the unirradiated cells; (iii) TNF-a is responsible for the production of free radicals; (iv) TNF-a-generated free radicals could cause genetic damage; and (v) the persistent NF-kB activation in the bystander cells mediates a concurrent development of acquired survival advantage of the bystander cells that favors the genetically aberrant cells to carry over to subsequent generation. Those cells could undergo clonal selection and neoplastic transformation. In the oncoming spring 2015 beam time, pro-tumorigenic cells will be exposed in the presence or absence of NF-kB or TNF-a blockers and allowed to undergo several doubling times. The cells after 30 – 40 passages will then be inoculated into nude mice and examined for tumor growth. One set of experiments with this tumor growth in nude mice has already been carried out. With the repeat experiments, the data will be combined for the next report.

Finally, in order to determine whether or not particle radiation could cause sustained oncogenic responses in vital organs and those responses show variability from organ to organ, we investigated, in normal C56BL/J6 mice, the transcriptional and translational oncogene target deregulation in response to iron ion (56Fe; 600 MeV/amu) radiation at a total dose of 0.8 Gy at a dose rate of 0.5 Gy/min. Delayed response in vital target organs including brain, gut, kidney, liver, lungs, small-intestine, large-intestine, spleen were investigated at day 30 post-irradiation. Tissue micro array and qPCR profiling revealed that high-LET radiation could activate the transcription of 19, 30, 37, 21, 78, 23, 63, and 8 onco-targets in brain, gut, kidney, large-intestine, liver, lungs, small-intestine, and spleen, respectively. Significantly, 9, 16, 9, 71, 1, 25, and 2 oncotargets were increased in gut, kidney, large-intestine, liver, lungs, small-intestine, and spleen. This will set a stage for the upcoming study on TNFRI and TNFRII knockout mice to understand the in vivo role of TNF-a in governing the NF-kB-triggered bystander effect. These mice are now under breeding processes. With the extension approved until October 31, 2015, these mice will be exposed to 56Fe ion (600 MeV/amu) in upcoming spring 2015 beam time.

Bibliography Type:	Description: (Last Updated: 04/11/2021)
Abstracts for Journals and Proceedings	Manickam K, Mohan S, Aravindan N, Blakely EA, Natarajan M. "Tumorigenic bystander effect after 56Fe-ion particle radiation exposure is mediated through sustained cross talk between pro-inflammatory cytokine, TNF-a and Rel family proteins." Radiation Quality Effects On Cancer Processes II-2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. Abstract #0342. , Jan-2015
Abstracts for Journals and Proceedings	Aravindan N, Natarajan M. "Tissue Dependent Molecular Basis for Neoplastic Transformation by 56Fe Ion Particle Radiation Exposure." Space Radiation Cancer I-NASA Human Research Program Investigators' Workshop. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. 2015 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 13-15, 2015. Abstract #0088. , Jan-2015
Abstracts for Journals and Proceedings	Natarajan M. "NF-kB Signaling in Radiation Response. Topical review talk." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. , Sep-2014
Abstracts for Journals and Proceedings	Krishnan M, Aravindan N, Mohan S, Gaudette C, Janardhanan P, Blakely E, Natarajan M. "Sustained TNF-a positive feedback loop in response to radiation exposure potentiates pro-tumorigenic to pre-malignant phenotype." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract #PS3-15. , Sep-2014
Abstracts for Journals and Proceedings	Durba MA, Aravindan S, Natarajan M, Ramraj S, Pandian V, Khan F, Herman T, Aravindan N. "Synthetic mono-ketone, EF24 affects autophagy in radio-resistant neuroblastoma cells." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract #PS3-63. , Sep-2014
Abstracts for Journals and Proceedings	Khan F, Masilamani R, Ramraj S, Pandian V, Aravindan S, Natarajan M. "Raspberry polyphenol targets cancer stem cells in residual pancreatic cancer after radiotherapy: A comprehensive in-vivo/Ex-vivo approach." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract #PS4-26. , Sep-2014
Abstracts for Journals and Proceedings	Ramraj S, Aravindan N, Natarajan M, Pandian V, Khan F, Herman T, Aravindan S. "High polarity polyphenol fractions from brown algae target radiation-induces tumor invasion and metastasis in therapy-resistant residual pancreatic cancer." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract #PS4-32. , Sep-2014

Abstracts for Journals and Proceedings	Pandian V, Natarajan M, Aravindan S, Khan F, Ramraj S, Herman T, Aravindan N. "Nitric oxide mediated inhibition of NF-kB arbitrates ER24-induced radio-sensitization in neuroblastoma tumor model." 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. 60th Annual Meeting of the Radiation Research Society, Las Vegas, Nevada, September 21-24, 2014. Abstract #PS4-33. , Sep-2014
Abstracts for Journals and Proceedings	Natarajan M. "Radiation and Carcinogenesis." Indian Medical Association-Continuing Medical Education (CME)-Annual Program, Tamilnadu, India, February 20, 2014. Indian Medical Association-Continuing Medical Education (CME)-Annual Program, Tamilnadu, India, February 20, 2014. , Feb-2014
Articles in Peer-reviewed Journals	Aravindan N, Aravindan S, Pandian V, Khan FH, Ramraj SK, Natt P, Natarajan M. "Acquired tumor cell radiation resistance at the treatment site is mediated through radiation-orchestrated intercellular communication." Int J Radiat Oncol Biol Phys. 2014 Mar 1;88(3):677-85. http://dx.doi.org/10.1016/j.ijrobp.2013.11.215 ; PubMed PMID: 24411622 , Mar-2014
Articles in Peer-reviewed Journals	Nair BC, Krishnan SR, Sareddy GR, Mann M, Xu B, Natarajan M, Hasty P, Brann D, Tekmal RR, Vadlamudi RK. "Proline, glutamic acid and leucine-rich protein-1 is essential for optimal p53-mediated DNA damage response." Cell Death Differentiation. 2014 Sep;21(9):1409-18. Epub 2014 May 2. http://dx.doi.org/10.1038/cdd.2014.55 ; PubMed PMID: 24786831; PubMed Central PMCID: PMC4131173 , Sep-2014
Articles in Peer-reviewed Journals	Aravindan S, Natarajan M, Ramraj SK, Pandian V, Khan FH, Herman TS, Aravindan N. "Abscopal effect of low-LET gamma-radiation mediated through Rel protein signal transduction in a mouse model of nontargeted radiation response." Cancer Gene Ther. 2014 Feb;21(2):54-9. Epub 2013 Dec 20. http://dx.doi.org/10.1038/cgt.2013.72 ; PubMed PMID: 24357814 , Feb-2014